

Lecture 7: More pointers & dynamic memory

CMPE I20: Computer Organization & Architecture

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What will be printed?

- Hint: draw the stack!

```
char b[] = {'C', 'M', 'P', 'E', '\0'};  
char *ptr_b = b;  
b[1] = 'O';  
ptr_b[3] = 'Y';  
char c = *b;
```

```
printf("Values: %c\n", ptr_b[1]);
```

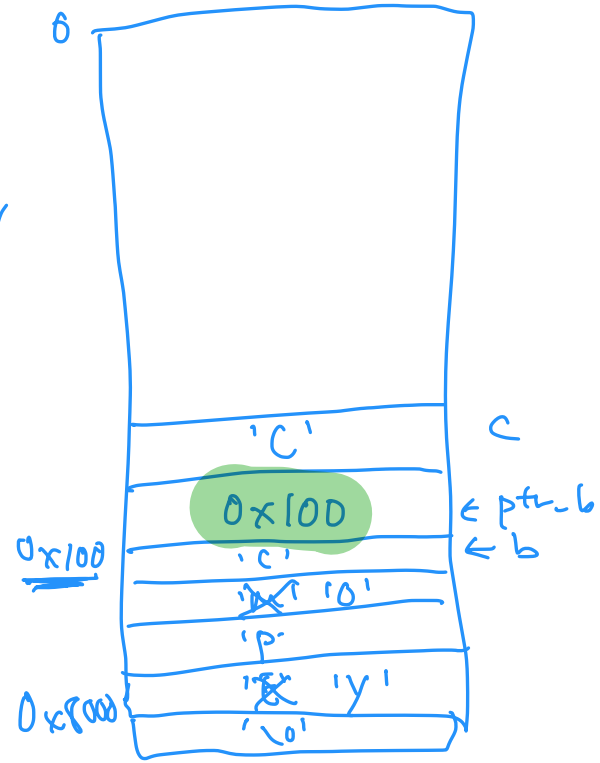
```
printf("Values: %c\n", b[1]);
```

```
printf("ptr_b = %s\n", ptr_b);
```

```
printf("b = %s\n", b);
```

```
printf("c = %c\n", c);
```

Null terminator



format specifiers
%.p : printing ptrs
%.s : string

Address or value?

& = get address of (some variable)

Bonus question: Does `&c == &num`? **NO**

```
int num = 42;
```

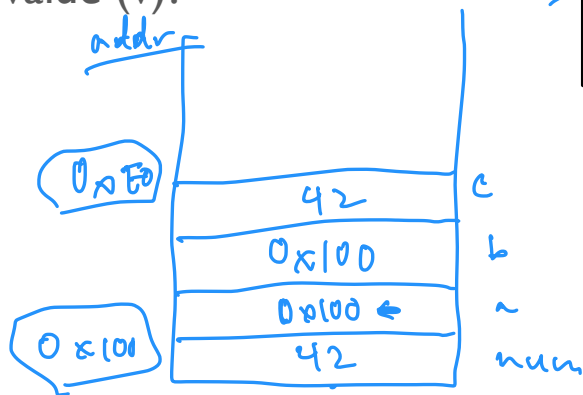
```
int *a = &num;
```

```
int *b = a;
```

```
int c = *a; // dereferencing a  
          ↳ 42
```

● Is this an address (a) or value (v)?

- a **a** ←
- *a ✓
- ○ c ✓
- b **a**
- *b ✓
- &c **a**



- A. v, a, v, v, a, v

B. v, a, a, a, v, a

C. a, v, **v**, a, v, a ✓

D. a, v, a, a, v, a

Demo

- `void swap_by_val(int a, int b)`
- `void swap_by_ptr(int *a, int *b)`

`swap_by_val(int a, int b) {`

`int temp = a;`

`a = b;`

`b = temp;`

`}`

A: Yes

B: NO

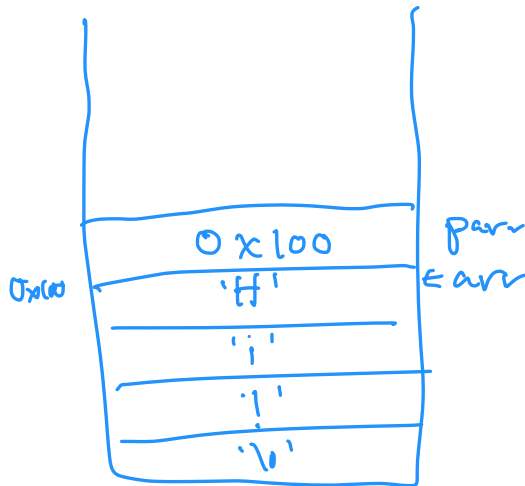
A pointer can also point to an array?

- An array is a region of memory allocated to a set of values of a specific data type
 - The name of an array corresponds to the address of the first element of an array

```
char arr[4] = {'H', 'i', '!', '\0'};  
char *parr = arr;
```

```
printf("Values: %c %c\n", arr[1], parr[1]);
```

```
printf("Addresses: %p %p\n", arr, parr);  
assert(arr == parr);
```



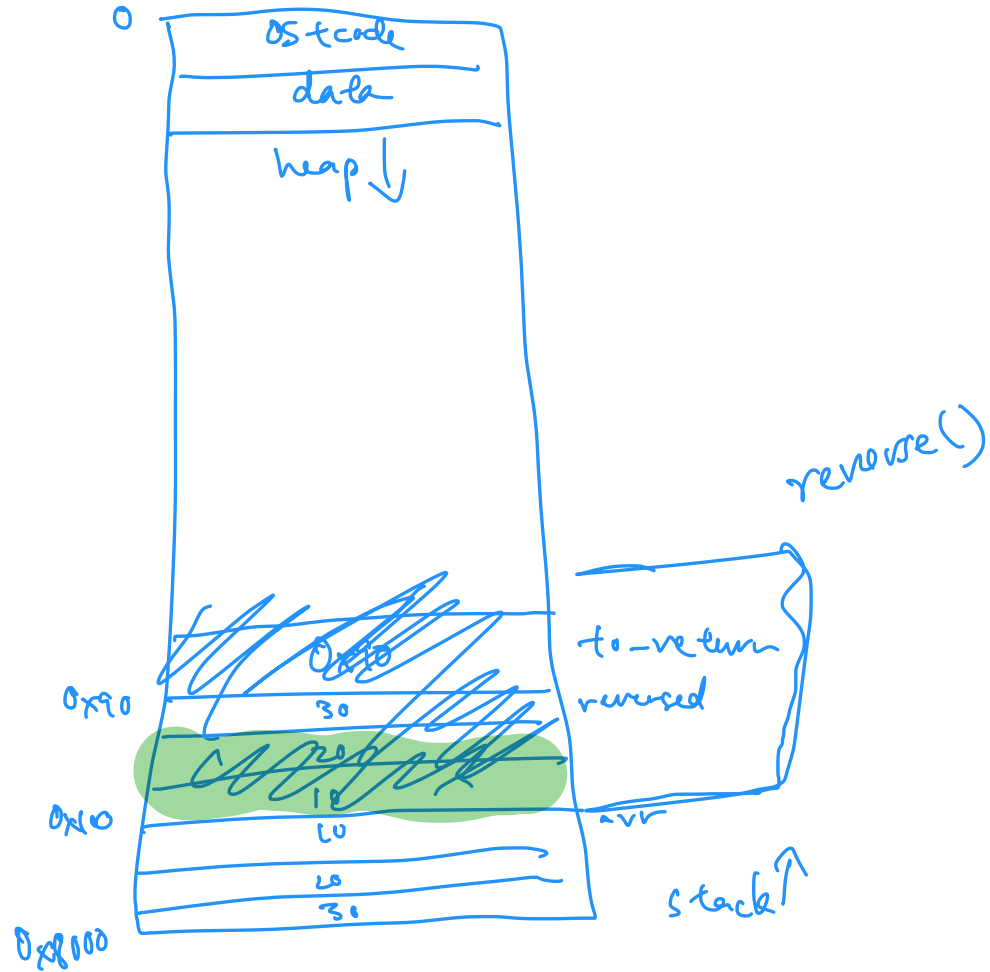
Demo

A: Yes

B: No

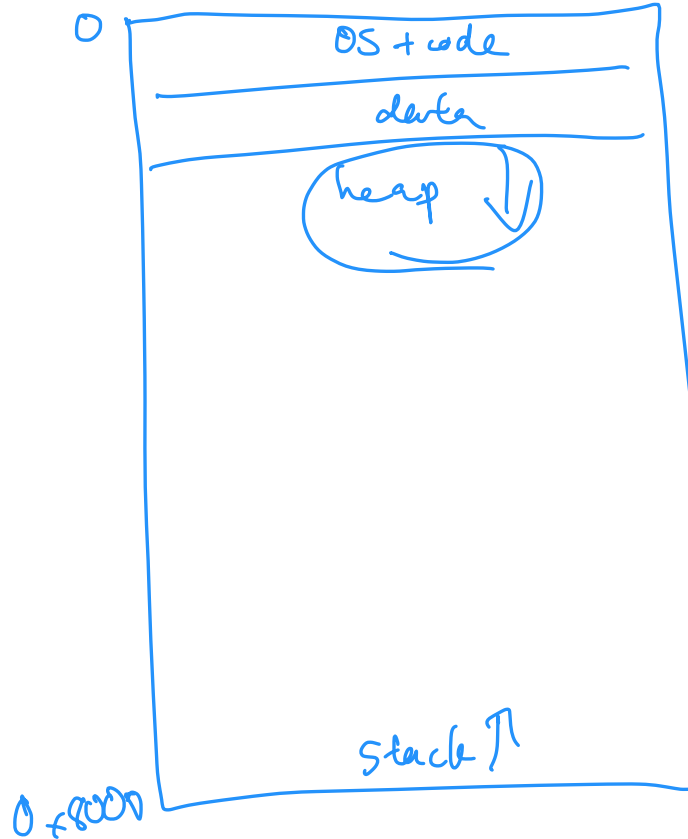
```
int *reverse(int arr[], int n) {  
    int reversed[n];  
    for (int i = 0; i < n; i++) {  
        reversed[i] = arr[n - i - 1];  
    }  
    int *to_return = reversed;  
    return to_return;  
}
```

```
int main() {  
    int arr[3] = {10, 20, 30};  
    int *reversed = reverse(arr, 3);  
    for (int i = 0; i < 3; i++) {  
        printf("%d\n", reversed[i]);  
    }  
}
```



What is the heap for?

program's address space



Heap memory

- Heap memory can be accessed from any function as long as we have pointers to it
- What if we want to create an array that persists after a function returns?
 - Use the heap!

How to request memory from the heap?

- malloc(num_bytes)
 - stands for "memory allocate"
 - returns the address of the first byte in the memory allocated
 - part of `stdlib.h`
- Example: variable-length array (length of the array changes over time)

`int *pa = malloc(3 * sizeof(int));`

bytes of a given data type

// can do whatever array things I want with pa

*pa[0]
pa+1*

stdlib.h

*sizeof(char) = 1 bytes
sizeof(int) = 4*

Demo

- `reverse_fix()`

What happens to heap memory after `malloc()`?

- Heap memory stays there forever so other functions can use it
 - Nice, but what happens when you're done?

We must free() the heap

- `free(void *ptr)`
 - Tells the heap to free memory that it allocated at this pointer
 - Now other processes can use it
- What happens if we do not free()?
 - **Memory leak!**
 - Performance degradation (slow down the computer)
 - Other processes cannot get the memory they need

What if my pointer points to nothing?

- A pointer points to address 0

```
int *p = 0x0;
```

```
printf("%d \n", *p); // segmentation fault!
```

addr 0 == NULL

What is a segmentation fault?

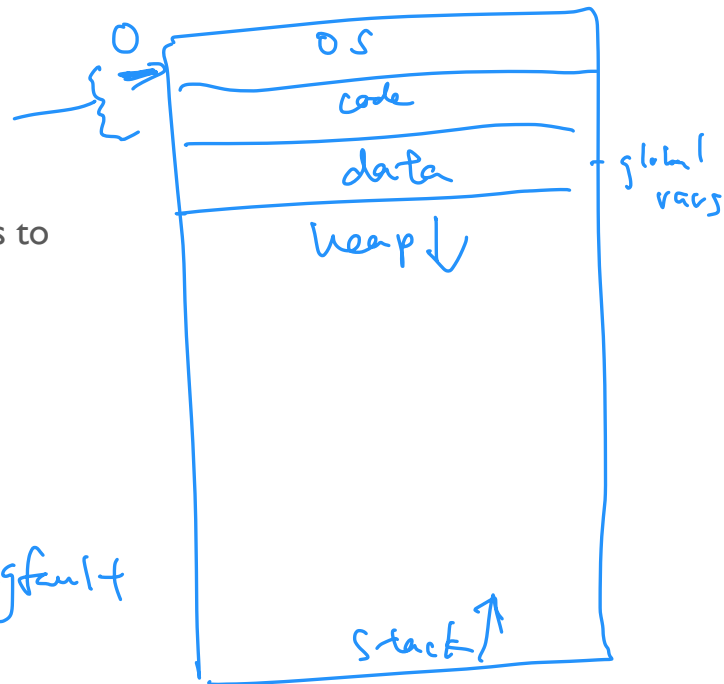
- **segfault**: when a program tries to access memory it is not allowed to access.

- When does this happen?

- Dereferencing a null pointer
- Dereferencing a pointer the program does not have access to

```
int *pnum = 42;  
printf("%i.p", pnum);  
42
```

*pnum → segfault



Should I panic when I segfault?

- No! Instead:
 - Look at the pointers you are working with
 - print them with %p
 - If it's 0:
 - likely dereferencing null pointer
 - If it's close to 0:
 - likely dereferencing a pointer you don't have access to
- How to fix?
 - Back trace and see how the pointer's value and/or address was changed